

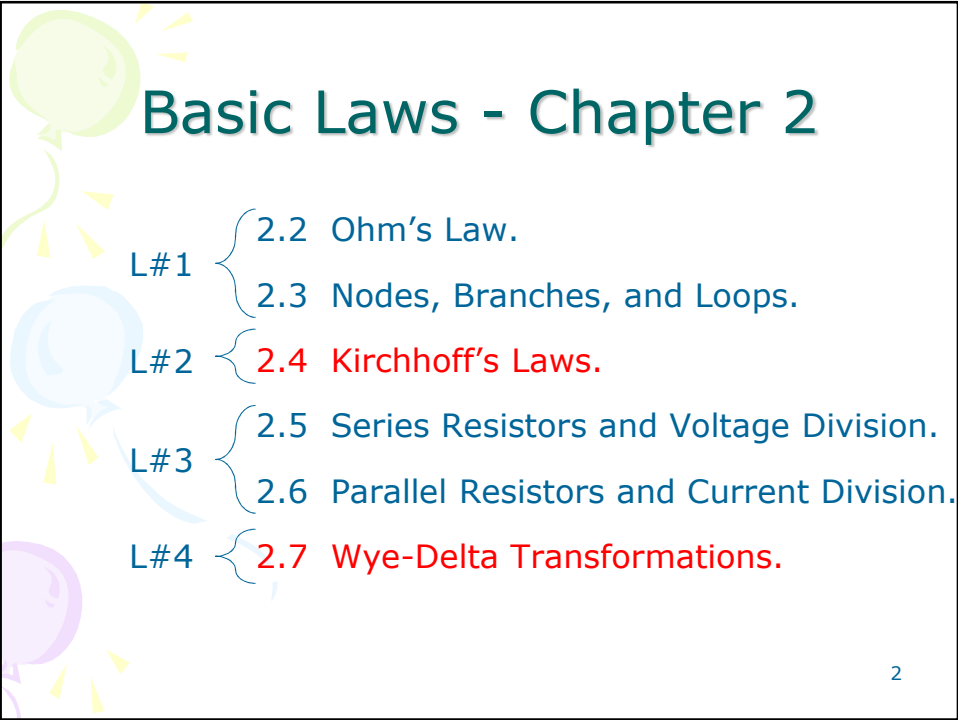
EBN 111/122
ELECTRICITY AND ELECTRONICS

Chapter 2

Basic Laws

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Basic Laws - Chapter 2

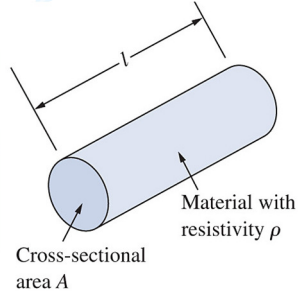
- L#1 { 2.2 Ohm's Law.
2.3 Nodes, Branches, and Loops.
- L#2 { 2.4 Kirchhoff's Laws.
- L#3 { 2.5 Series Resistors and Voltage Division.
2.6 Parallel Resistors and Current Division.
- L#4 { 2.7 Wye-Delta Transformations.

2

Chapter 2: Lecture #1

2.2 Ohms Law: $V = IR$

- Ohm's law states that the voltage V across a resistor is directly proportional to the current I flowing through the resistor.
- Resistance R : Physical property of a material to resist the flow of current.



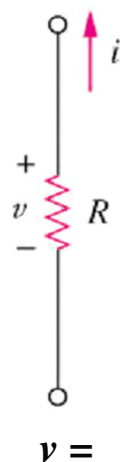
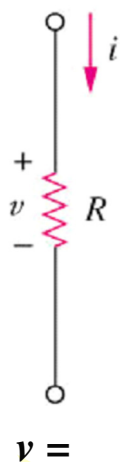
$$R = \rho \frac{l}{A} = \frac{V}{I}$$



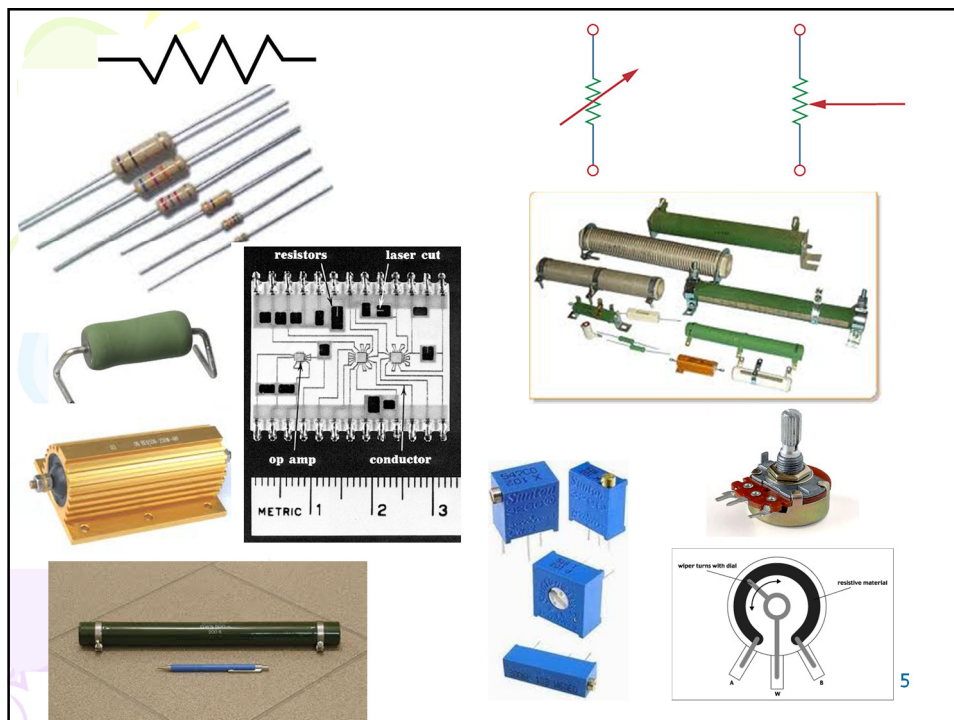
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2.2 Ohms Law: $V = IR$

Passive Sign Convention



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2.2 Ohms Law

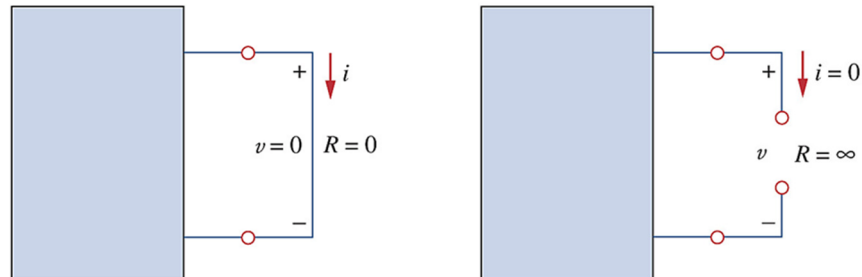
Resistivity ρ & Resistance R

Material	Resistivity ($\Omega\cdot m$)	Usage
Copper	1.72×10^{-8}	Conductor
Gold	2.45×10^{-8}	Conductor
Carbon	4×10^{-5}	Semi-conductor
Silicon	6.4×10^2	Semi-conductor
Paper	1×10^{10}	Insulator
Teflon	3×10^{12}	Insulator

$$R = \rho \frac{l}{A} = \left[\Omega \cdot m \frac{m}{m^2} \right] = [\Omega]$$

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2.2 Ohms Law **Resistance R**



$$R = \frac{V}{I}$$

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2.2 Ohms Law **Conductance G**

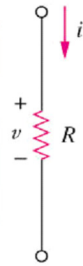
- Conductance is the ability of an element to conduct electric current; it is the reciprocal of resistance R and is measured in mhos [U] or Siemens [S].

$$G = \frac{1}{R} = \frac{i}{v} [S]$$

$$i = Gv$$

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2.2 Ohms Law

Power dissipated in a Resistor

$$p = vi$$

$$v = iR$$

$$i = v/R$$

$$= (iR)i = i^2R$$

$$= v \frac{v}{R} = \frac{v^2}{R}$$

MUCHO IMPORTANTE:

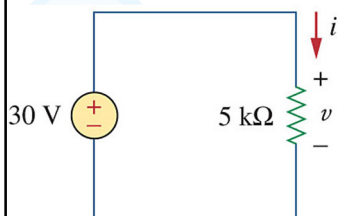
Since i^2 , v^2 & R is always ≥ 0 :

P in a resistor is **ALWAYS ≥ 0** (positive)

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Example 2-2

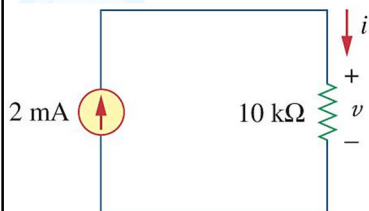
In the circuit shown, calculate the current i , the conductance G and the power p .



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PP 2-2

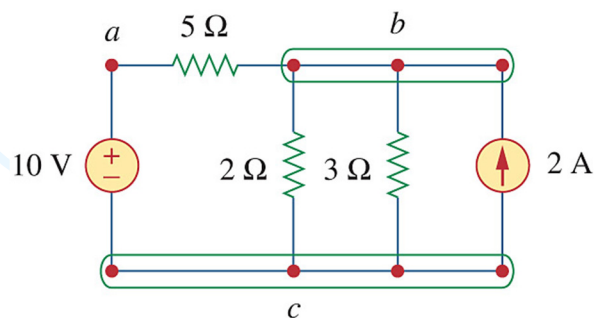
For the circuit shown, calculate the voltage v , conductance G and power p .



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2.3 Nodes, Branches and Loops

- A **branch** represents a single element such as a voltage source or a resistor.
- A **node** is the point of connection between two or more branches.
- A **loop** is any closed path in a circuit.



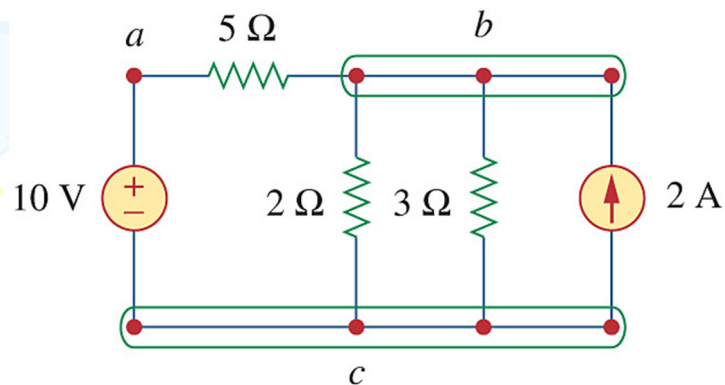
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2.3 Nodes, Branches and Loops

BRANCHES

"A single element such as a voltage source or a resistor."

How many branches?



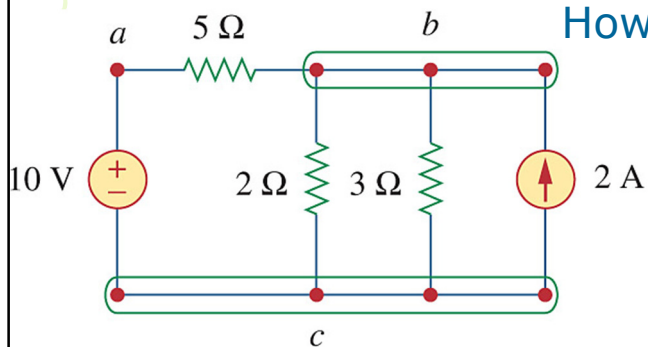
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2.3 Nodes, Branches and Loops

NODES

"The point of connection between two or more branches."

How many nodes?



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2.3 Nodes, Branches and Loops

LOOPS

"Any closed path in a circuit."

How many loops?

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2.3 Nodes, Branches and Loops

INDEPENDENT LOOPS

"A loop that contains at least one branch which is not a part of any other independent loop."

"A loop that cannot be formed by combining / merging any of the other loops."

Independent sets leads to independent equations

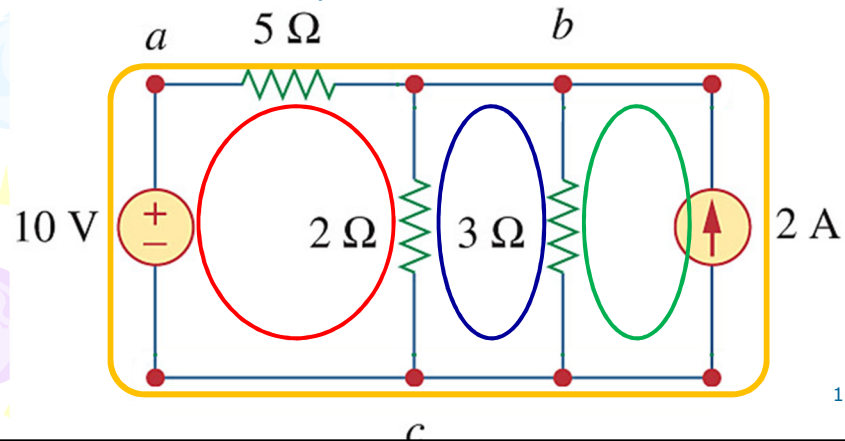
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2.3 Nodes, Branches and Loops

When is it NOT an independent set?

If any loop can be formed by merging any of the other loops.

Dependent set 1:

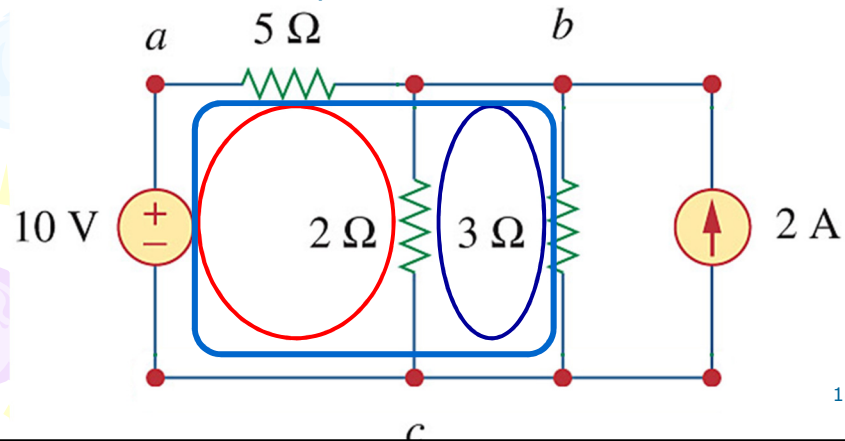


2.3 Nodes, Branches and Loops

When is it NOT independent?

If any loop can be formed by merging any of the other loops.

Dependent set 2:



2.3 Nodes, Branches and Loops

When is it NOT independent?

If any loop can be formed by merging any of the other loops.

Dependent set 3:

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2.3 Nodes, Branches and Loops

Fundamental theorem of network topology

A network with b branches, n nodes, and l independent loops will satisfy the fundamental theorem of network topology:

$$b = l + n - 1$$

# branches	=	# ind-loops	+	# nodes	-	1

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2.3 Nodes, Branches and Loops

Series and Parallel

SERIES

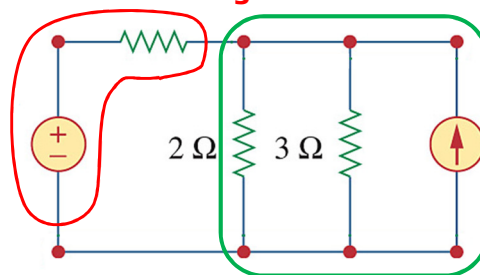
Two (2) elements are in series if:

- They exclusively share a single node
- $\therefore$ carry the **same current**.

PARALLEL

Two or more elements are in parallel if:

- They are connected to the same two nodes
- $\therefore$ have the **same voltage**



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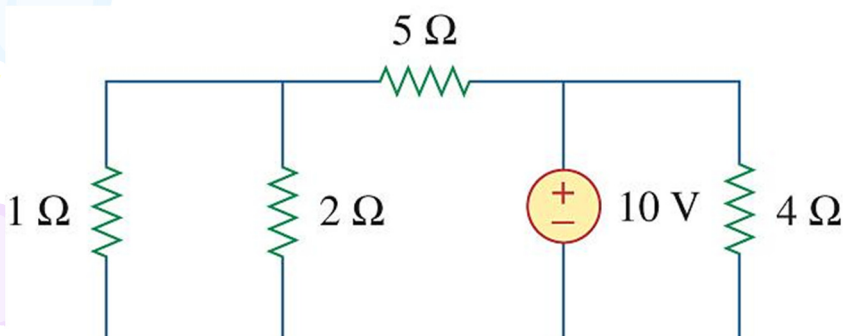
PP 2-4

How many branches, nodes and independent loops does the circuit have?

Branches =

Nodes =

Loops =



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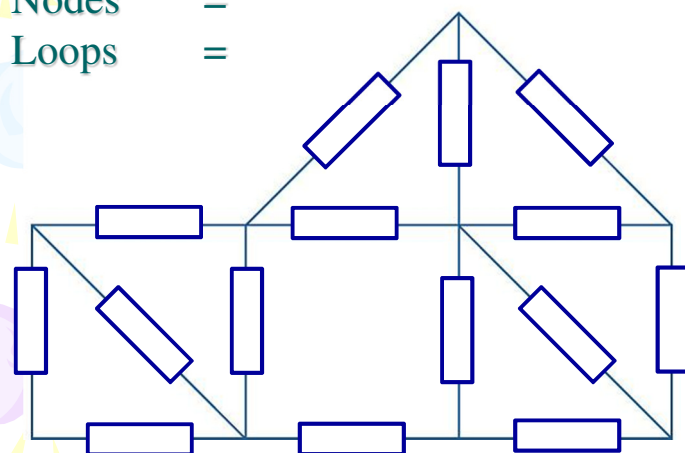
Problem 2.5

Find the number of nodes, branches and independent loops.

Branches =

Nodes =

Loops =



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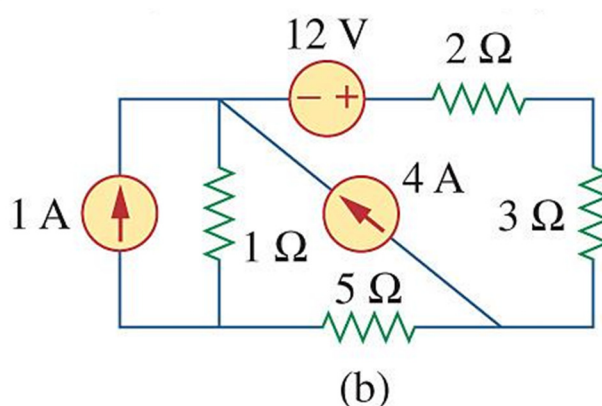
Problem 2.7b

Find the number of nodes, branches and independent loops.

Branches =

Nodes =

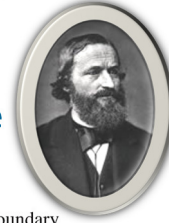
Loops =



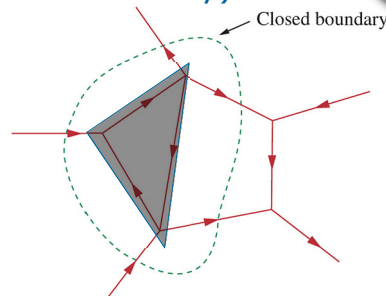
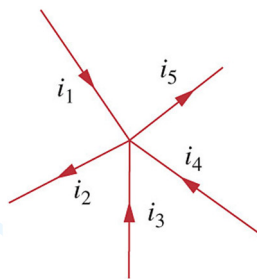
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Chapter 2: Lecture #2

2.4 Kirchhoff's Laws



- Kirchhoff's **current law (KCL)** states that the algebraic sum of currents entering and leaving a node (or a closed boundary) is zero.



Mathematically
$$\sum_{n=1}^N i_n = 0$$

N = Number of branches connected to the node.

i_n = n^{th} current entering or leaving.

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2.4 Kirchhoff's Laws (KCL)

"What goes in, must come out"

Current entering a node: **ADD +**
 Current leaving a node: **SUBTRACT -**

$$i_1 + (-i_2) + i_3 + i_4 + (-i_5) = 0$$

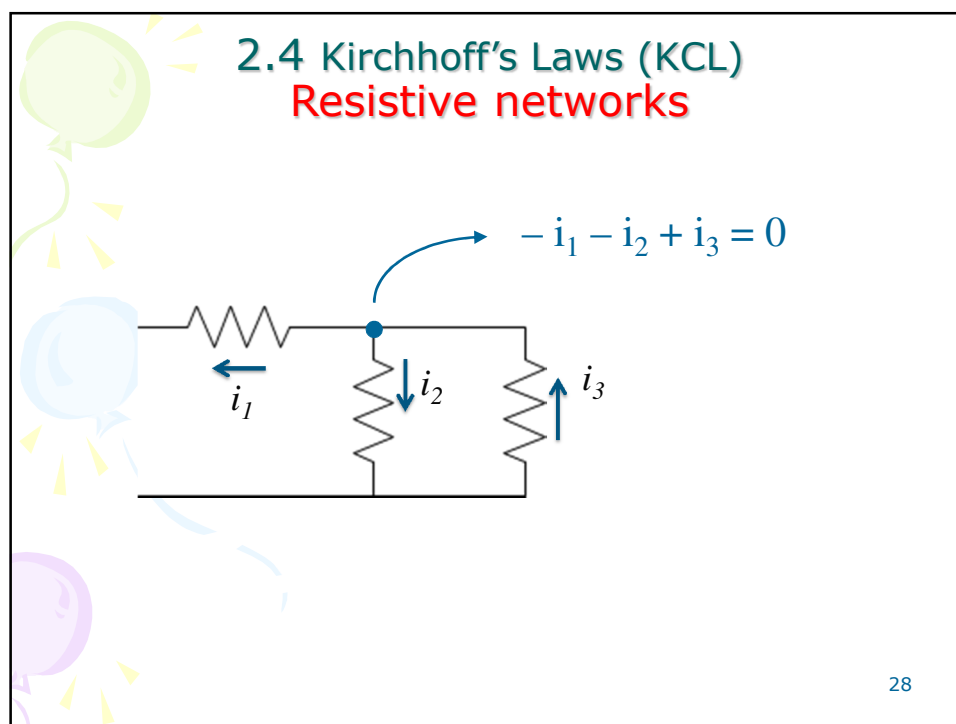
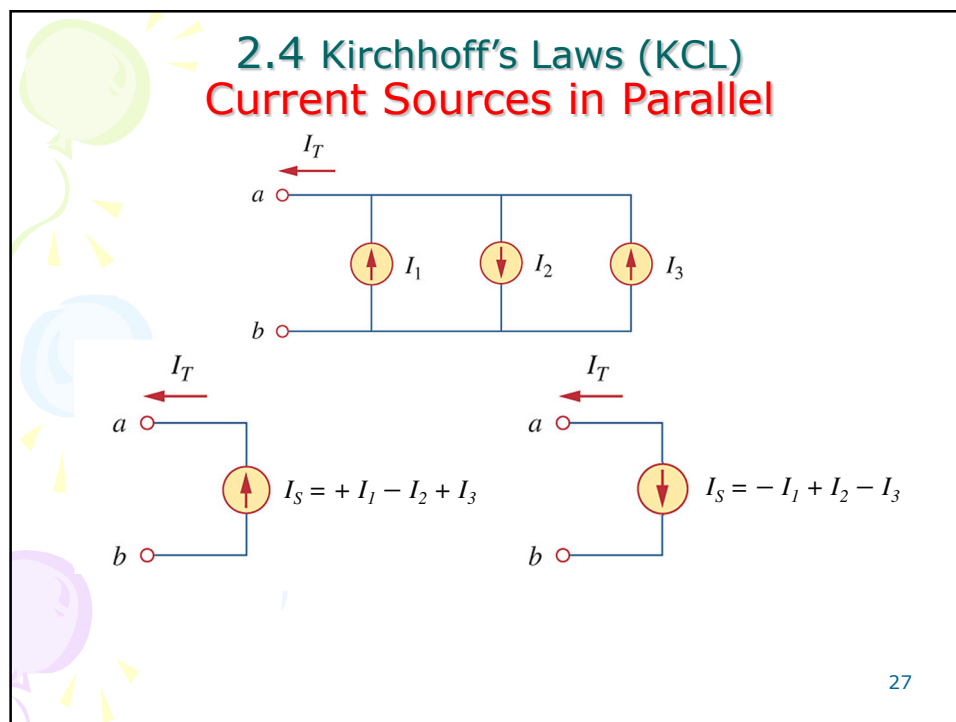
Entering = Leaving

$$i_1 + i_3 + i_4 = i_2 + i_5$$

Closed Boundary (Super Node):

$$-i_1 - i_2 - i_3 + i_4 + i_5 = 0$$

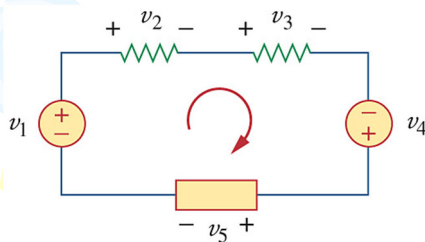
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2.4 Kirchhoff's Laws (KVL)

"What goes up, must come down"

- Kirchhoff's **voltage law (KVL)** states that the algebraic sum of all voltages around a closed path (or loop) is zero.



Convention:

- Always go clock wise
- In at +: **ADD**
- In at -: **SUBTRACT**

$$-v_1 + v_2 + v_2 - v_4 + v_5 = 0$$

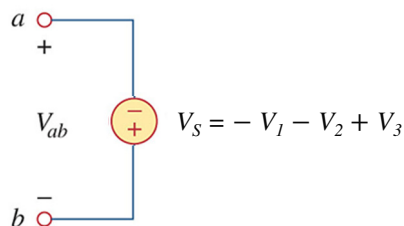
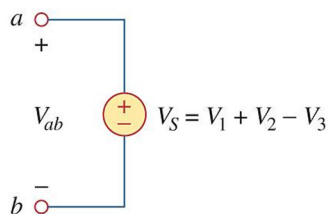
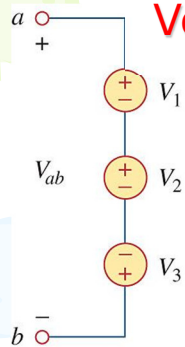
Mathematically,

$$\sum_{m=1}^M v_m = 0$$

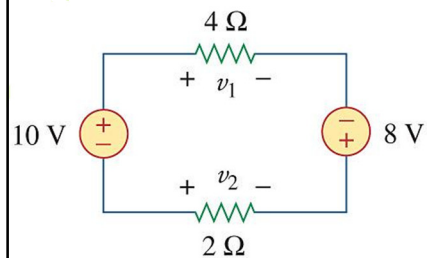
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2.4 Kirchhoff's Laws (KVL)

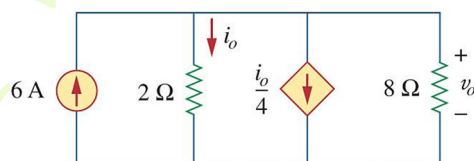
Voltage sources in series



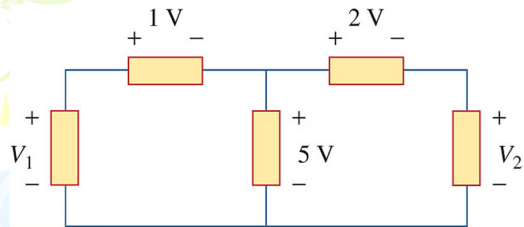
30

PP 2-5Find i , v_1 and v_2 .

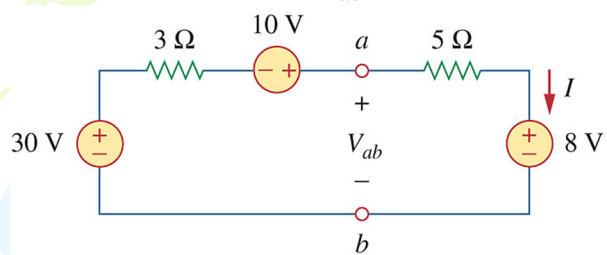
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PP 2-7Find current i_o and voltage v_o .

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Problem 2-11Calculate V_1 and V_2 .

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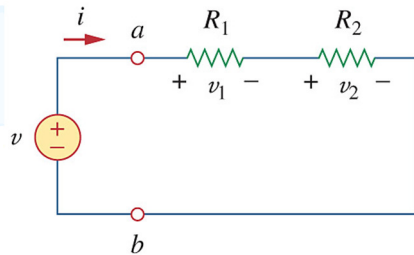
Problem 2-18Determine I and V_{ab} .

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Chapter 2: Lecture #3

2.5 Series Resistors & Voltage Division

- **Series:** Two or more elements are in series if they are cascaded or connected sequentially and consequently carry the same current.



$$v_1 = iR_1 \text{ or } i = \frac{v_1}{R_1}$$

$$v_2 = iR_2 \text{ or } i = \frac{v_2}{R_2}$$

$$\text{KVL: } -v + v_1 + v_2 = 0$$

$$v = v_1 + v_2$$

$$= i(R_1 + R_2)$$

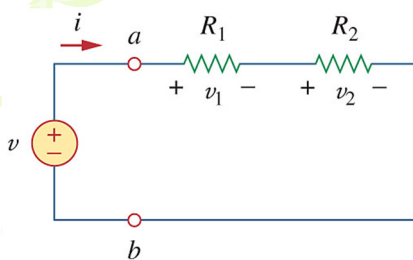
$$= iR_{eq}$$

$$i = \frac{v}{R_1 + R_2}$$

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2.5 Series Resistors & Voltage Division

VOLTAGE DIVISION



$$v_1 = iR_1$$

$$v_2 = iR_2$$

$$i = \frac{v}{R_1 + R_2}$$

Substitute i into v_1 and v_2 :

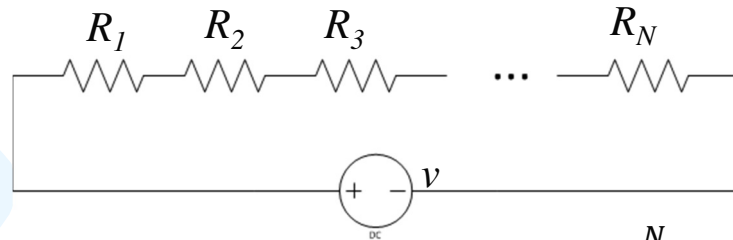
$$v_1 = \frac{v}{R_1 + R_2} R_1 = \frac{R_1}{R_1 + R_2} v = \frac{R_1}{R_{eq}} v$$

$$v_2 = \frac{v}{R_1 + R_2} R_2 = \frac{R_2}{R_1 + R_2} v = \frac{R_2}{R_{eq}} v$$

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2.5 Series Resistors & Voltage Division

R_{eq} for more than 2 R's



$$R_{eq} = R_1 + R_2 + R_3 + \dots + R_N = \sum_{n=1}^N R_n$$

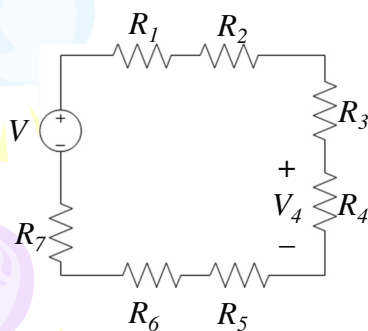
$$v_n = \frac{R_n}{R_1 + R_2 + \dots + R_N} v$$

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2.5 Series Resistors & Voltage Division

R_{eq} for more than 2 R's

$$R_{eq} = R_1 + R_2 + R_3 + R_4 + R_5 + R_6 + R_7 = \sum_{n=1}^7 R_n$$



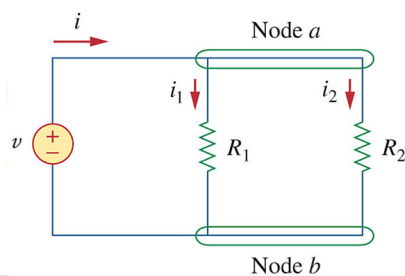
$$V_4 = \frac{R_4}{R_{eq}} V$$

$$V_6 = \frac{R_6}{R_{eq}} V$$

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2.6 Parallel Resistors & Current Division

- Parallel:** Two or more elements are in parallel if they are connected to the same two nodes and consequently **have the same voltage across them**.



$$i_1 = v/R_1 \quad i_2 = v/R_2$$

$$v = i_1 R_1 = i_2 R_2$$

$$\text{KCL: } i - i_1 - i_2 = 0$$

$$i = i_1 + i_2$$

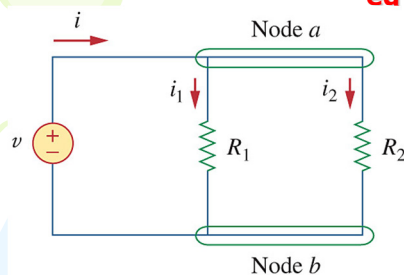
Substitute i_1 and i_2 :

$$i = \frac{v}{R_1} + \frac{v}{R_2} = v \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = \frac{v}{R_{eq}}$$

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2.6 Parallel Resistors & Current Division

R_{eq} for 2 R's



$$i = v \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = \frac{v}{R_{eq}}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

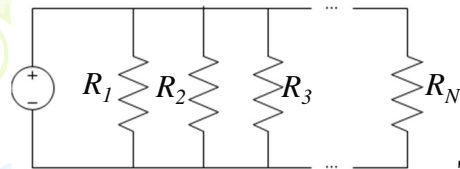
$$= \frac{R_1 + R_2}{R_1 R_2}$$

$$R_{eq} =$$

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2.6 Parallel Resistors & Current Division

R_{eq} for more than 2 R's



If

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

Then does it mean that:

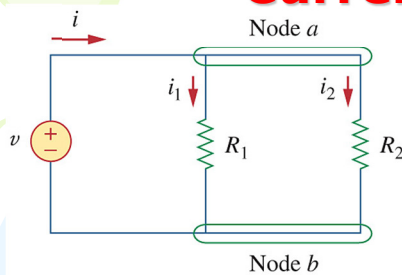
$$R_{eq} = \frac{R_1 R_2 R_3 \cdots R_N}{R_1 + R_2 + R_3 + \cdots + R_N}$$

$$\frac{1}{R_{eq}} =$$

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2.6 Parallel Resistors & Current Division

Current Division



$$v = iR_{eq} = i \frac{R_1 R_2}{R_1 + R_2}$$

$$v = i_1 R_1 = i_2 R_2$$

Substituting v :

$$i_1 R_1 = i \frac{R_1 R_2}{R_1 + R_2}$$

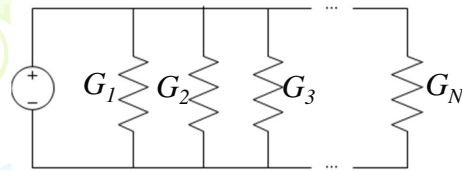
$$i_1 = i \frac{R_2}{R_1 + R_2}$$

$$i_2 = i$$

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2.6 Parallel Resistors & Current Division

Current Division – more than 2 R's



$$G_n = \frac{1}{R_n}$$

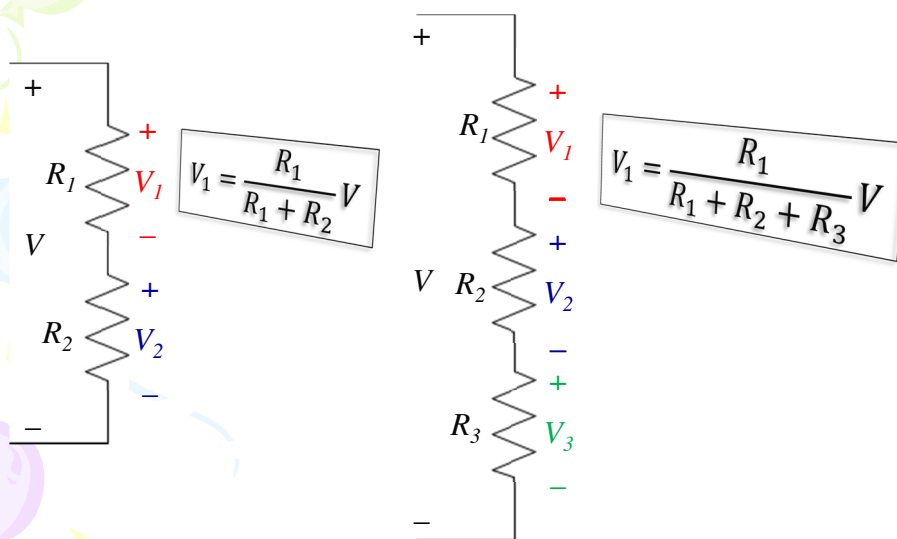
$$G_{eq} = G_1 + G_2 + G_3 + \cdots + G_N = \sum_{n=1}^N G_n$$

$$i_n = \frac{G_n}{G_1 + G_2 + G_3 + \cdots + G_N} i$$

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2.6 Parallel Resistors & Current Division

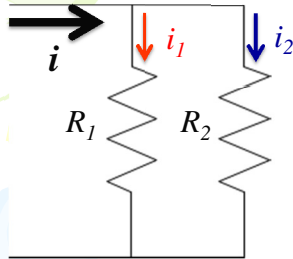
Voltage Division - Summary



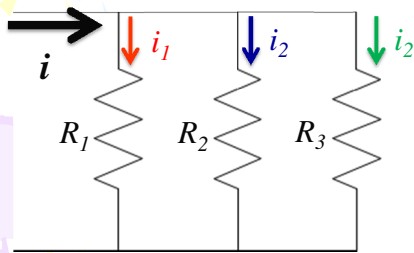
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2.6 Parallel Resistors & Current Division

Current Division - Summary



$$i_1 = i \frac{R_2}{R_1 + R_2}$$



$$i_1 = ?$$

$$i_1 = i \frac{R_2 R_3}{R_1 + R_2 + R_3}$$



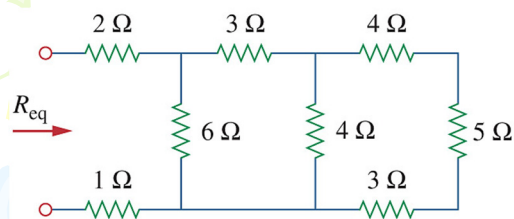
$$i_1 = i \frac{G_1}{G_1 + G_2 + G_3}$$



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PP 2-9

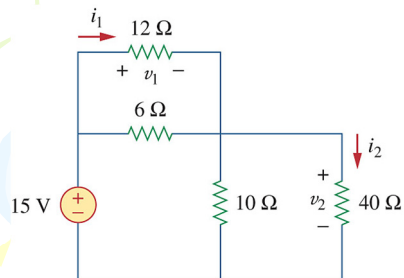
Determine R_{eq}



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PP 2-12

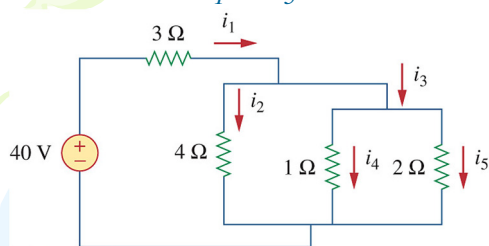
Find v_1 and v_2 in the circuit. Also calculate i_1 and i_2 and the power dissipated in the $12\ \Omega$ and $40\ \Omega$ resistors.



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Problem 2-31

Determine i_1 to i_5 .

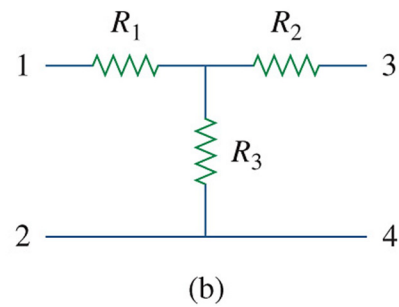
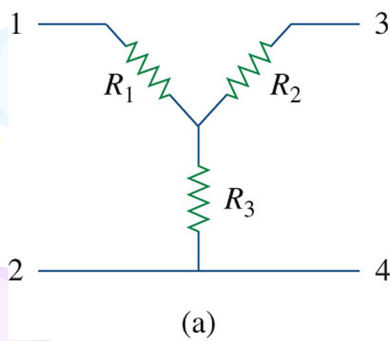


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Chapter 2: Lecture #4

2.7 Wye-Delta Transformations

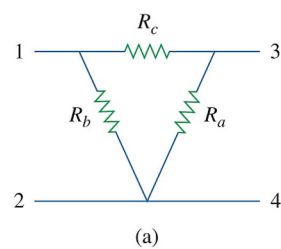
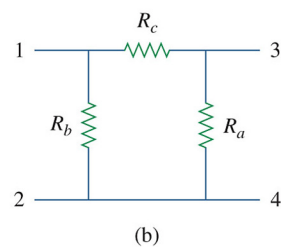
WYE / Y or T or Star Network



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2.7 Wye-Delta Transformations

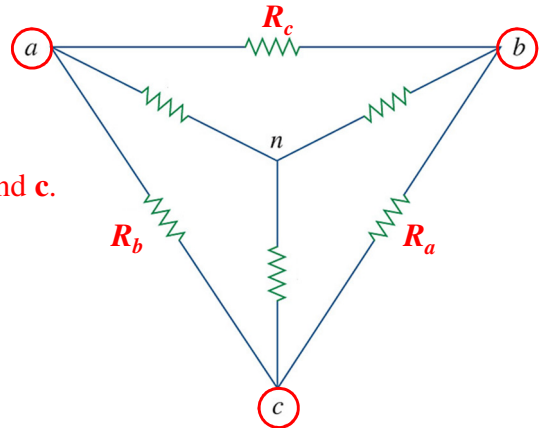
Δ or π Network

 Δ  π

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2.7 Wye-Delta Transformations

Y inside a Δ



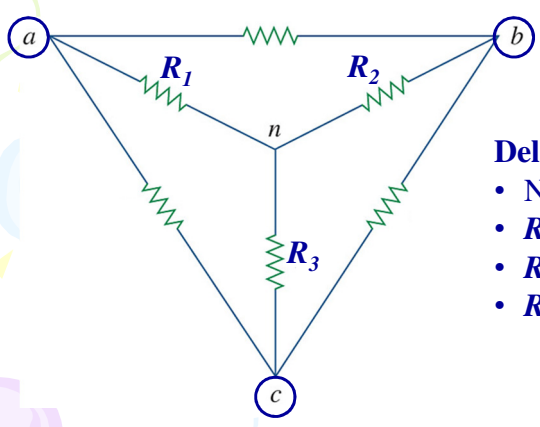
Delta Δ : R_a, R_b, R_c .

- Name the nodes **a**, **b**, and **c**.
- R_a is opposite node **a**
- R_b is opposite node **b**
- R_c is opposite node **c**

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2.7 Wye-Delta Transformations

Y inside a Δ



Delta Y: R_1, R_2, R_3 .

- Name the nodes **a**, **b**, and **c**.
- R_1 is connected to node **a**
- R_2 is connected to node **b**
- R_3 is connected to node **c**

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2.7 Wye-Delta Transformations

Y inside a Δ

Delta $\rightarrow$ Wye

$Y \leftarrow \Delta$

Product of adjacent Δ R's

$$R_1 = \frac{R_b R_c}{(R_a + R_b + R_c)}$$

$$R_2 = \frac{R_c R_a}{(R_a + R_b + R_c)}$$

$$R_3 = \frac{R_a R_b}{(R_a + R_b + R_c)}$$

Wye $\rightarrow$ Delta

$\Delta \leftarrow Y$

All possible products of Y R's 2@t

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

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2.7 Wye-Delta Transformations

Special Case: Balanced Y and Δ

$$R_a = R_b = R_c = R_\Delta$$

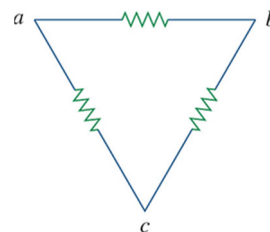
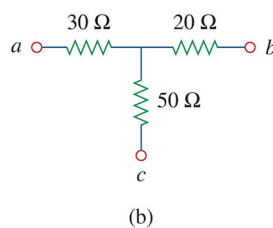
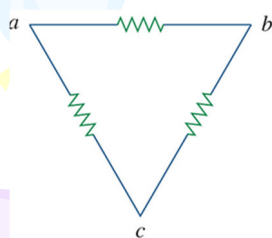
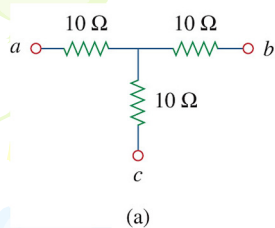
and

$$R_1 = R_2 = R_3 = R_Y$$

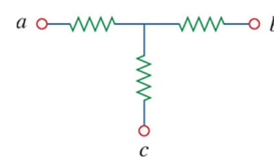
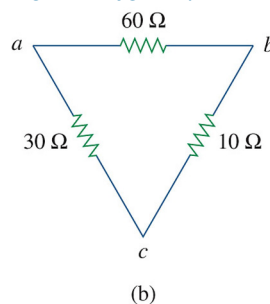
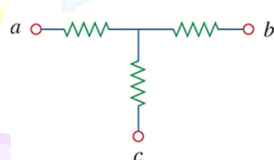
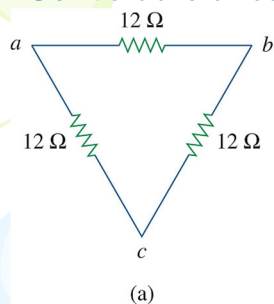
then:

$$R_Y = \frac{R_\Delta}{3} \text{ or } R_\Delta = 3R_Y$$

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Problem 2-48Convert the circuits from Y to Δ :

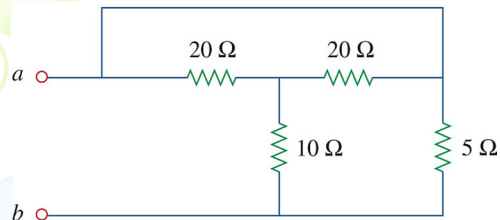
55

Problem 2-49Convert the circuits from Δ to Y :

56

Problem 2-44

Determine R_{eq} between terminals a and b :



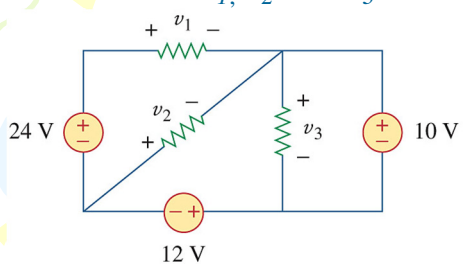
57

Chapter 2: Lecture #5

Additional Problems

Problem 2-17

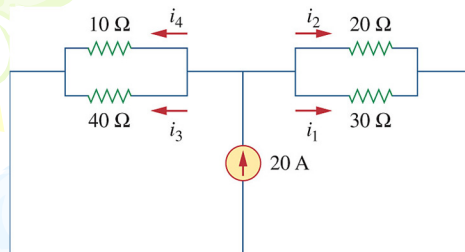
Determine v_1 , v_2 and v_3 in the circuit.



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Problem 2-32

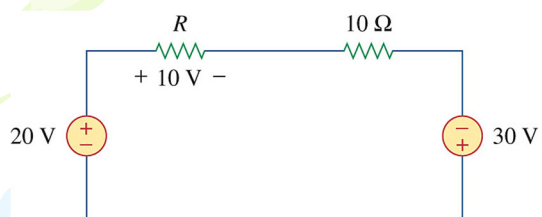
Determine i_1 , i_2 , i_3 and i_4 in the circuit.



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Problem 2-37

Find R for the circuit.



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Problem 2-55Calculate I_o in the circuit.